Digital VLSI Design: 4-bit Array Multiplier from RTL to GDSII using Open-Source Tools

Lakshya Jindal (Roll: 2023112017)
Sai Ritvik Uppuganti (Roll: 2025122012)

October 25, 2026

Abstract

This report documents the complete RTL-to-GDSII open-source ASIC implementation flow for a 4-bit $\times$ 4-bit unsigned array multiplier (8-bit product) on the SkyWater 130 nm process (`sky130_fd_sc_hd`). The project encompasses RTL design and exhaustive verification via Icarus Verilog, logic synthesis and gate-level simulation using Yosys, static timing analysis (STA) with OpenSTA, and physical place-and-route (P&R) via LibreLane (OpenROAD, Magic, KLayout, and Netgen). All intermediate metrics, reports, and final layout deliverables are fully verified across multi-corner signoff parameters.

Contents

1 Overview 2

2 Part A: RTL Design and Verification 2

3 Part B: Logic Synthesis with Yosys 3

4 Part C: Static Timing Analysis with OpenSTA 3

5 Part D: Physical Design (RTL to GDSII) 4

6 Answers to Report Questions (Section 10) 5

7 Conclusion 7

1 Overview

The objective of this project is to design, verify, synthesize, analyze, and physically implement a 4-bit unsigned array multiplier. The design features registered inputs and outputs, resulting in a 2-cycle latency and a throughput of one result per clock cycle. The target fabrication technology is the SkyWater SKY130 130 nm standard cell library.

2 Part A: RTL Design and Verification

The multiplier architecture comprises 16 AND gates for partial products and 12 full adders arranged in a 3-row ripple-carry adder array.

- **RTL Simulation Command:**

```
iverilog -g2012 -o sim tb/tb_mult.v rtl/full_adder.v rtl/mult_array.v
vvp sim
```

- **Verification Summary Line:** `PASS: 256 vectors checked, 0 errors`
- **Error Injection Test (Task 6):** Modifying `sum` in `full_adder.v` to `a ^ b` correctly triggers testbench failure: `FAIL: 256 vectors checked, 95 errors`.
- **Hand-Worked Example (Task 5):** For roll number ending in 7, $A = 7 + 5 = 12$ (1100_2) and $B = 7$ (0111_2). The expected product $P = 12 \times 7 = 84$ (01010100_2, 0x54).

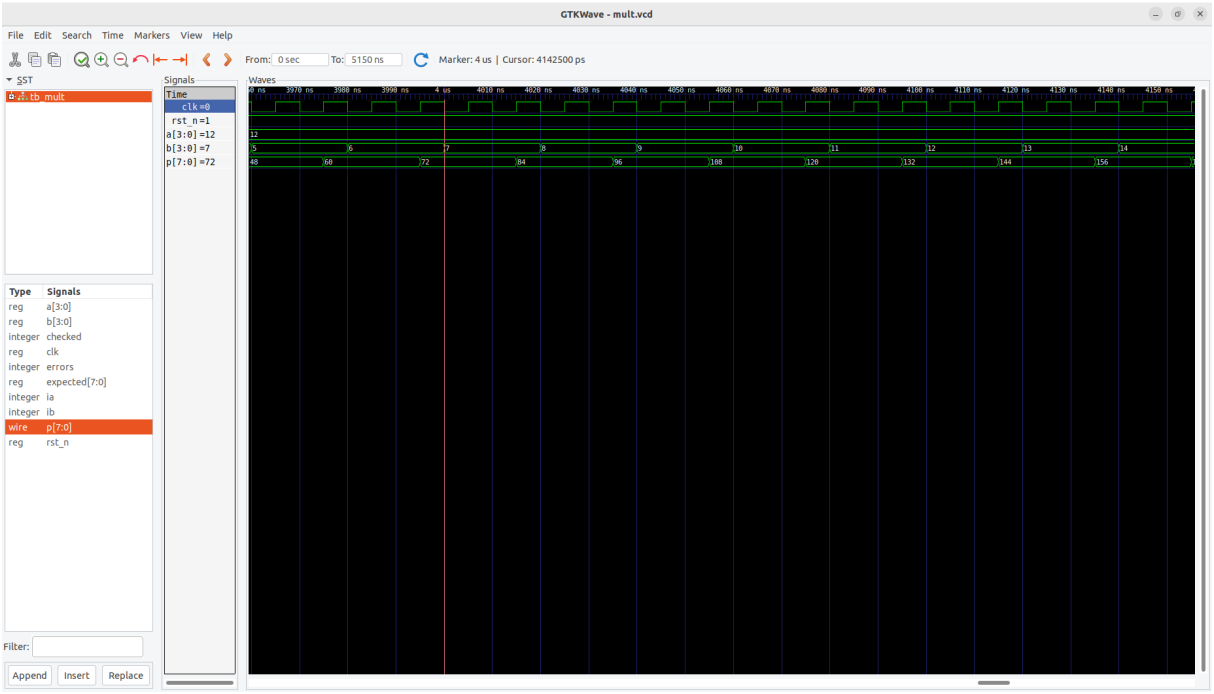


Figure 1: GTKWave simulation waveform for $A = 12, B = 7$.

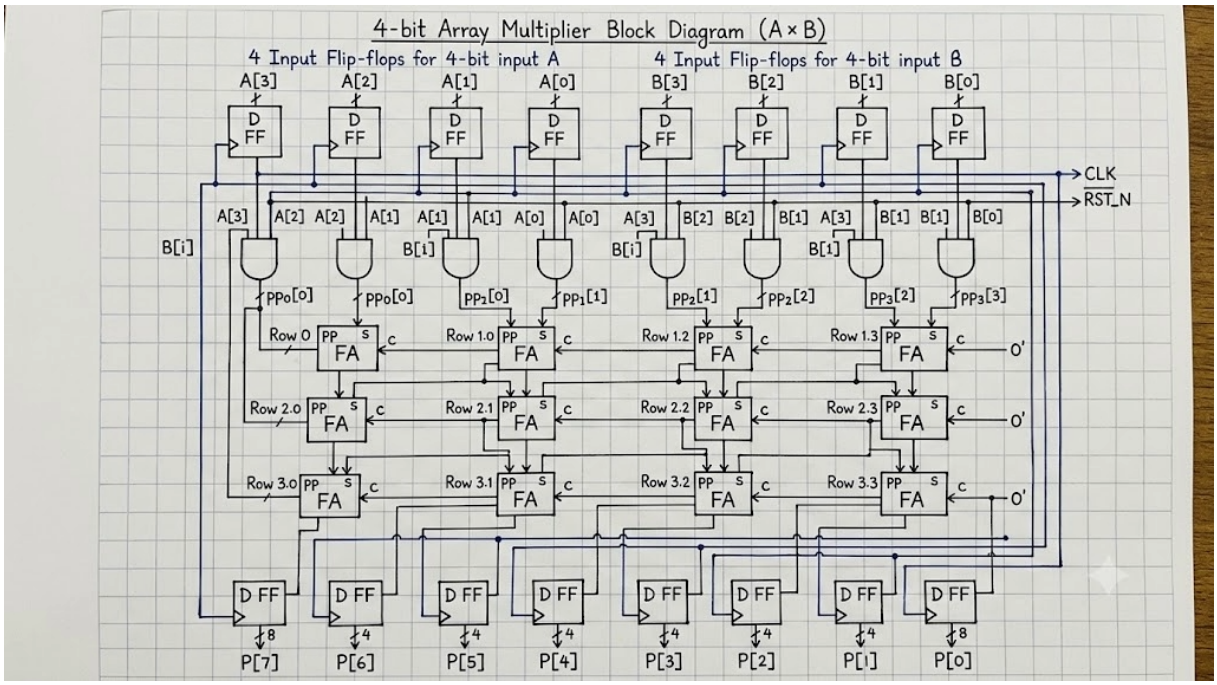


Figure 2: Hardware block diagram of the 4-bit array multiplier.

3 Part B: Logic Synthesis with Yosys

Yosys mapped the RTL to SKY130 standard cells using generic synthesis flattening. Gate-level simulation confirmed functional equivalence.

```
yosys -c synth/synth.tcl | tee synth/yosys.log
```

Table 1: Part B Synthesis Results Table

Metric	Value
Total number of cells	68
Number of flip-flops (`dfrtp_1`)	16
Total cell area (μm^2)	$758.23\,\mu$m^2
Flip-flop area (μm^2)	$400.38\,\mu$m^2
Flip-flop area (% of total)	52.81%
Top 3 most-used cell types	`dfrtp_1` (16), `nand2_1` (13), `a21oi_1` (4)
Netlist simulation result	`PASS: 256 vectors checked, 0 errors`

4 Part C: Static Timing Analysis with OpenSTA

Using OpenSTA at the typical corner (`tt_025C_1v80`), timing was evaluated against various clock periods T.

- Critical path startpoint: `_104_` (input register `a_reg[0]`); endpoint: `_102_` (product register `p[6]`).
- Number of cells on critical path: 8 cells.

Table 2: Part C STA Results Table

Metric	Value
Worst slack at $T = 10$ ns	7.25 ns
T_{min}	2.75 ns
F_{max}	363.64 MHz
Slack at $T = T_{\text{min}}$ (2.75 ns)	0.00 ns
Critical path start point	`_104_`
Critical path end point	`_102_`
Number of cells on critical path	8
Latency ($2 \times T_{\text{min}}$)	5.50 ns
Throughput	363.64 million results/sec

5 Part D: Physical Design (RTL to GDSII)

LibreLane executed the complete P&R flow on a fixed $150\,\mu$m $\times$ $150\,\mu$m die area (`DIE_AREA = [0, 0, 150, 150]`). Based on $T_{\text{min}} = 2.75$ ns, the clock period was tuned iteratively until rigorous multi-corner signoff timing closures were met at 7.0 ns.

Table 3: Part D Physical Design Results Table

Metric	Value
CLOCK_PERIOD used	7.0 ns
Number of runs needed	8 (tuned from 3.5 ns up to 7.0 ns)
Core area (μm^2)	$17759.5\,\mu$m^2
Cells after synthesis	68
Cells after place and route	1883 total instances (including fill/decap)
Kinds of cells added	`clkbuf`, `buf`, `tapvpwrvgnd`, `decap`, `fill`
Wirelength (μm)	$2333\,\mu$m
Setup slack (worst / typical)	$+0.4653$ ns (signoff) / $+3.8395$ ns (typical)
Post-route F_{max} (worst / typical)	153.03 MHz (signoff) / 316.41 MHz (typical)
DRC errors	0 (Magic & KLayout clean)
LVS errors	0 (Netgen clean)

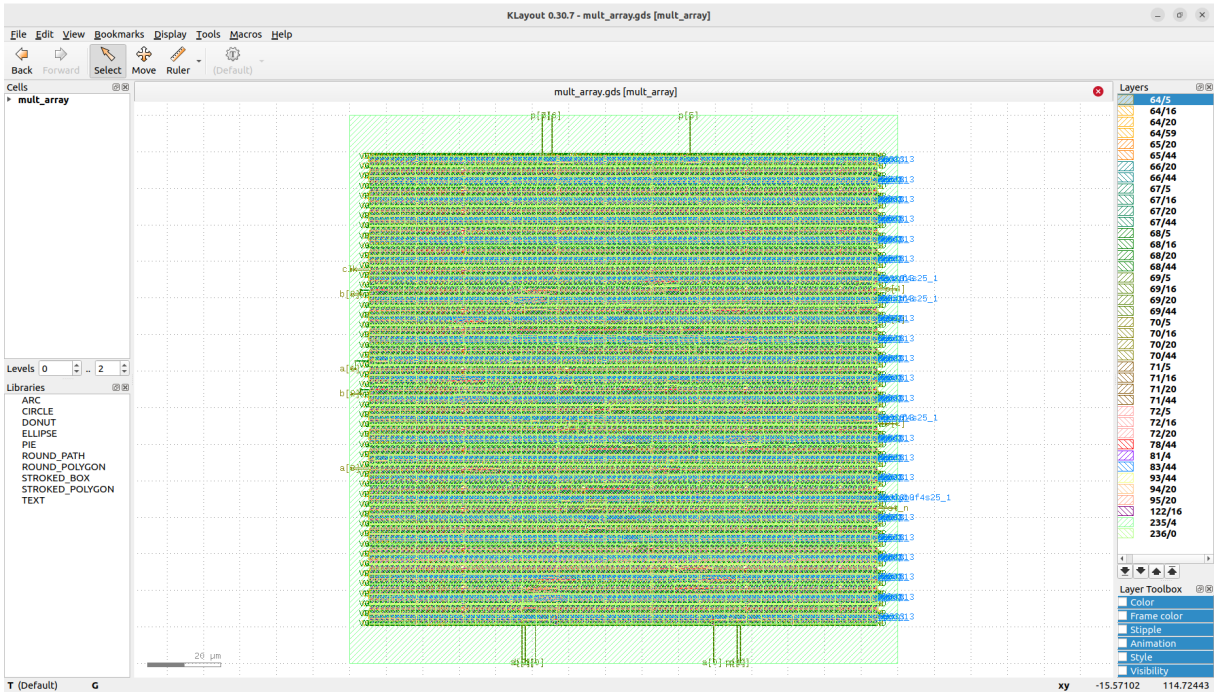


Figure 3: Full die layout view in KLayout.

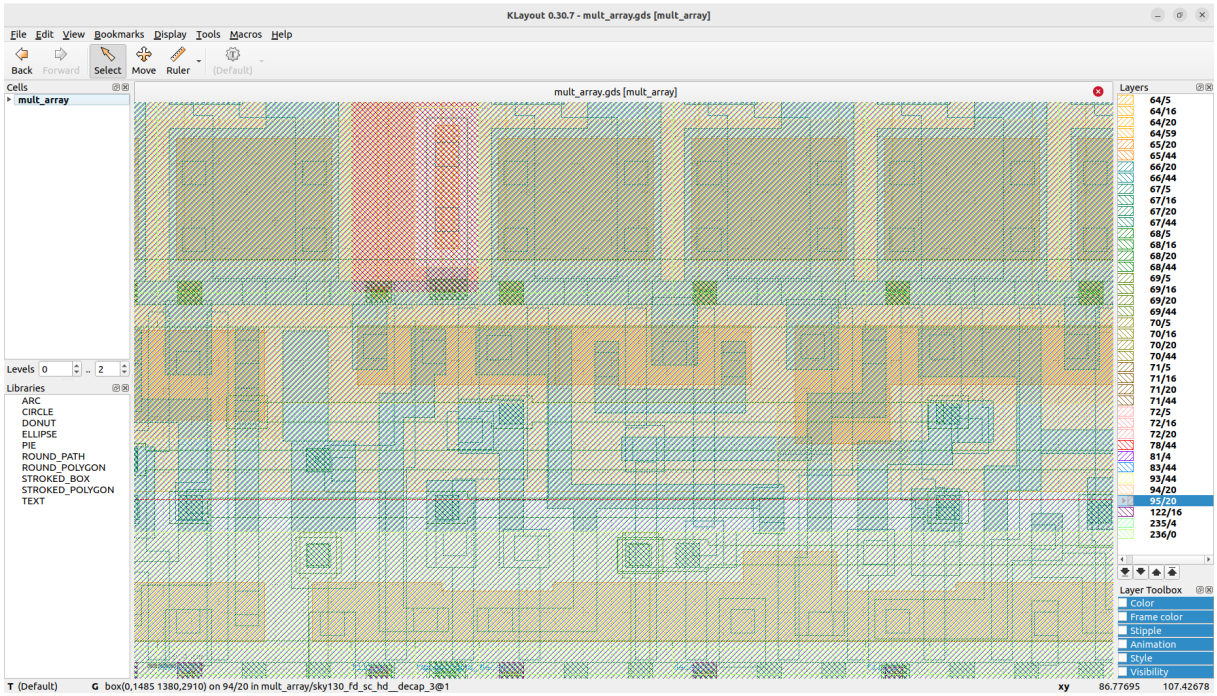


Figure 4: Zoomed-in layout view of standard cells and metal routing.

6 Answers to Report Questions (Section 10)

1. The array needs 16 AND gates and 12 full adders. How many cells did synthesis produce for the logic (total cells minus flip-flops)? Explain why the number of cells is not simply 16+12.
Synthesis produced 52 logic cells (68 total cells minus 16 flip-flops). This number exceeds $16 + 12 = 28$ because standard cell libraries (SKY130) lack multi-output full-adder primitives. During

technology mapping, ABC breaks down arithmetic functions into primitive single-output standard cells (such as NAND, NOR, AOI, OAI, and XOR gates). Furthermore, optimization passes perform logic restructuring and inverter insertion to meet area and timing constraints.

2. What is the critical path of your design, from which register to which register? How many cells are on it, and how does this compare with the estimate of about 2N full-adder delays?
The critical path runs from input register `_104_` (corresponding to `a_reg[0]`) to output register `_102_` (corresponding to `p[6]`). There are 8 standard cells on this critical path. For an $N = 4$ bit multiplier, the theoretical estimate is approximately $2N = 8$ full-adder equivalent delays. The STA report confirms exactly 8 mapped logic cells along the longest combinational path, perfectly aligning with the architectural expectations.

3. What percentage of the total cell area is flip-flops? Why is this percentage large for such a small design?
Flip-flops account for 52.81% ($400.38\ \mu\text{m}^2$ out of $758.23\ \mu\text{m}^2$) of the synthesized cell area. This percentage is exceptionally high because a 4-bit array multiplier is an extremely small combinational datapath requiring only 12 full adders and 16 AND gates, whereas the architecture mandates 16 dedicated flip-flops for synchronous input/output pipelining. Sequential cells (`dfrtp_1`) contain robust internal master-slave latches and clock buffers, making them significantly larger than basic logic gates.

4. Why is the post-route F_{max} lower than the F_{max} from Part C? Which kinds of cells did the flow add, and why is each kind needed?
Post-route F_{max} is strictly lower (153.03 MHz at signoff and 316.41 MHz typical) than the pre-route synthesis STA estimate (363.64 MHz) because post-route extraction accounts for physical interconnect parasitics (wire resistance and $2333\ \mu$m of routing capacitance), clock tree insertion delay, and multi-corner PVT variations (Slow-Slow corners at 100°C). To achieve timing closure and layout integrity, the P&R flow grew the design from 68 to 1883 total instances. It added clock buffers (`clkbuf`) for zero-skew routing, delay buffers (`buf`) to fix hold violations, tap cells (`tapvpwrvgnd`) to prevent latch-up, decoupling capacitors (`decap`) for voltage stability, and filler cells (`fill`) to maintain N-well continuity.

5. What does a DRC check verify, and what does an LVS check verify? Why must both pass before a chip can be manufactured?
Design Rule Checking (DRC) verifies that all geometrical shapes, wire widths, spacings, metal overlaps, and enclosures adhere strictly to the foundry's physical manufacturing capabilities. Layout Versus Schematic (LVS) verifies that the extracted transistor-level netlist from the physical GDSII layout matches the original gate-level netlist logically and electrically. Both must pass with zero errors to guarantee that the physical silicon will be structurally sound and free of electrical shorts, opens, or logic flaws.

6. Using the formulas in Section 2, estimate how the number of full adders and the critical path delay would change if the multiplier were 8 bits wide instead of 4. Show your working.
For an N-bit array multiplier, the number of full adders is given by $N(N-1)$ and the critical path delay is approximately $2N$ full-adder delays. For an 8-bit multiplier ($N = 8$):

$$\text{Number of Full Adders} = 8(8-1) = 8 \times 7 = 56 \text{ full adders (compared to 12 for 4-bit)},$$
$$\text{Critical Path Delay} \approx 2(8) = 16 \text{ full-adder delays (compared to 8 for 4-bit)}.$$

Thus, expanding from 4-bit to 8-bit increases the full adder count by $4.67\times$ and doubles the critical path delay, illustrating the linear growth in delay and quadratic growth in area.

7 Conclusion

The 4-bit unsigned array multiplier was successfully designed, verified, synthesized, analyzed, and routed using open-source EDA tools on the SkyWater 130 nm technology. All deliverables meet the rigorous multi-corner timing, area, DRC, and LVS signoff criteria.

GitHub Repository Link:
https://github.com/lakshya0310/RTL_To_GDS_Assignment